

J12 PWM_B

JST-SH 1x8 · 3.3 V logic

1	NC						
2	GND						
3	NC						
4	ESC telemetry	ONE_WIRE	LPUART7_TX		GPIO4_I031	GPIO_EMC_31	
5	PWM5	PWM_5	FLEXPWM4_PwMA01	FLEXIO1_FLEXIO02	GPIO4_I002	GPIO_EMC_02	
6	PWM6	PWM_6	FLEXPWM4_PwMB01	FLEXIO1_FLEXIO03	GPIO4_I003	GPIO_EMC_03	
7	PWM7	PWM_7	FLEXPWM4_PwMA02	FLEXIO1_FLEXIO04	GPIO4_I004	GPIO_EMC_04	
8	PWM8	PWM_8	FLEXPWM4_PwMB02	FLEXIO1_FLEXIO05	GPIO4_I005	GPIO_EMC_05	

Pin 4: LPUART7 in single-wire mode, receive only

J13 PWM_A

JST-SH 1x8 · 3.3 V logic

1	NC						
2	GND						
3	NC						
4	ESC telemetry	ONE_WIRE	LPUART7_TX		GPIO4_I031	GPIO_EMC_31	
5	PWM1	PWM_1	FLEXPWM2_PwMA00	FLEXIO1_FLEXIO06	GPIO4_I006	GPIO_EMC_06	
6	PWM2	PWM_2	FLEXPWM2_PwMB00	FLEXIO1_FLEXIO07	GPIO4_I007	GPIO_EMC_07	
7	PWM3	PWM_3	FLEXPWM2_PwMA01	FLEXIO1_FLEXIO08	GPIO4_I008	GPIO_EMC_08	
8	PWM4	PWM_4	FLEXPWM2_PwMB01	FLEXIO1_FLEXIO09	GPIO4_I009	GPIO_EMC_09	

Pin 4: LPUART7 in single-wire mode, receive only

J6 SPI (OSD)

JST-GH 1x8 (BM08B-GHS-TBT), top entry · 3.3 V logic

1	5V out	5V					
2	SPI SCK	OSD_SCK	LPSP12_SCK	GPIO3_I007	GPIO_SD_B1_07		
3	SPI MISO	OSD_MISO	LPSP12_SDI	GPIO3_I009	GPIO_SD_B1_09		
4	SPI MOSI	OSD_MOSI	LPSP12_SDO	GPIO3_I008	GPIO_SD_B1_08		
5	SPI CS	OSD_CS	LPSP12_PCS0	GPIO3_I006	GPIO_SD_B1_06		
6	Data_ready	OSD_DRDY		GPIO3_I010	GPIO_SD_B1_10		
7	nRST	OSD_nRESET		GPIO3_I011	GPIO_SD_B1_11		
8	GND						

J3 DEBUG

JST-SH 1x10 · DS-009 debug full · 3.3 V logic

PX4: console, /dev/ttyS0 (LPUART6)

1	VREF 3.3V	3V3							
2	UART_TX	CONSOLE_TX	LPUART6_TX	GPIO4_I025	GPIO_EMC_25				
3	UART_RX	CONSOLE_RX	LPUART6_RX	GPIO4_I026	GPIO_EMC_26				
4	SWDIO		JTAG_TMS	GPIO1_I006	GPIO_AD_B0_06				
5	SWCLK	SWDCLK	JTAG_TCK	GPIO1_I007	GPIO_AD_B0_07				
6	SW0		ARM_TRACE_SW0	GPIO2_I013	GPIO_B0_13				
7	NC								GPIO1 in DS-009, not wired
8	NC								GPIO2 in DS-009, not wired
9	nRST		POR_B						chip reset input, active low
10	GND								

J4 GPS

JST-GH 1x10 (BM10B-GHS-TBT), top entry · DS-009 GPS · 3.3 V logic

PX4: GPS, /dev/ttyS1 (LPUART2)

1	5V out	5V							
2	UART_TX	GPS_TX	LPUART2_TX	GPIO1_I018	GPIO_AD_B1_02				
3	UART_RX	GPS_RX	LPUART2_RX	GPIO1_I019	GPIO_AD_B1_03				
4	I2C_SCL	EXT_SCL	LPI2C1_SCL	GPIO1_I016	GPIO_AD_B1_00				
5	I2C_SDA	EXT_SDA	LPI2C1_SDA	GPIO1_I017	GPIO_AD_B1_01				
6	Safety switch	ARM	XBAR1_INOUT10	GPIO2_I012	GPIO_B0_12				
7	Safety LED	GPS_LED	FLEXPWM1_PwMB03	GPIO3_I025	GPIO_EMC_39				
8	3.3V out	3V3							
9	Buzzer	BUZZER_EXT	FLEXPWM1_PwMA00	GPIO4_I023	GPIO_EMC_23				via FET Q1
10	GND								

J2 POWER

Molex Click-Mate 1x6 (5024430670) · 3.3 V logic on SCL and SDA

1	5V in	5V					
2	5V in	5V					
3	I2C_SCL	PM_SCL	LPI2C3_SCL	GPIO4_I022	GPIO_EMC_22		
4	I2C_SDA	PM_SDA	LPI2C3_SDA	GPIO4_I021	GPIO_EMC_21		
5	GND						
6	GND						

J5 CAN

JST-GH 1x4, top entry · DS-009 CAN

120 Ω termination: fitted

1	5V out	5V					
2	CAN_H	CAN_H	CANH	TJA1462 CAN FD transceiver	TXD	CAN3_TX	GPIO_EMC_36
3	CAN_L	CAN_L	CANL		RXD	CAN3_RX	GPIO_EMC_37
4	GND						

J10 I2C

JST-GH 1x4, top entry · DS-009 I2C · 3.3 V logic

1	5V out	5V					
2	I2C_SCL	EXT_SCL	LPI2C1_SCL	GPIO1_I016	GPIO_AD_B1_00		
3	I2C_SDA	EXT_SDA	LPI2C1_SDA	GPIO1_I017	GPIO_AD_B1_01		
4	GND						

J11 RC

JST-GH 1x4, top entry · 3.3 V logic

PX4: RC, /dev/ttyS6 (LPUART8)

1	5V out	5V	500 mA fuse				
2	UART_TX	RC_TX	LPUART8_TX	GPIO1_I026	GPIO_AD_B1_10		
3	UART_RX	RC_RX	LPUART8_RX	GPIO1_I027	GPIO_AD_B1_11		
4	GND						

P1 ETH

JST-GH 1x2, top entry

P1 Main board → TJA1103 100BASE-T1 PHY → i.MX RT1064 Soc

1	T1 pair -	ETH_TRX_M	via TJA1103
2	T1 pair +	ETH_TRX_P	via TJA1103

J7 AUX

JST-GH 1x6 (BM06B-GHS-TBT), top entry · 3.3 V logic

PX4: AUX, /dev/ttyS3 (LPUART4)

1	5V out	5V					
2	UART_TX	AUX_TX	LPUART4_TX	FLEXPWM1_PwMA03	XBAR1_INOUT14	GPIO2_I016	GPIO_B1_00
3	UART_RX	AUX_RX	LPUART4_RX	FLEXPWM1_PwMB03	XBAR1_INOUT15	GPIO2_I017	GPIO_B1_01
4	UART_CTS	AUX_CTS	XBAR1_IN25	FLEXCAN2_RX		GPIO1_I015	GPIO_AD_B0_15
5	UART_RTS	AUX_RTS	XBAR1_IN24	FLEXCAN2_TX		GPIO1_I014	GPIO_AD_B0_14
6	GND						

J8 TELEM1

JST-GH 1x6 (BM06B-GHS-TBT), top entry · DS-009 TELEM · 3.3 V logic

PX4: TELEM1, /dev/ttyS2 (LPUART3)

1	5V out	5V					
2	UART_TX	TELEM_TX	LPUART3_TX	GPIO1_I022	GPIO_AD_B1_06		
3	UART_RX	TELEM_RX	LPUART3_RX	GPIO1_I023	GPIO_AD_B1_07		
4	UART_CTS	TELEM_CTS	LPUART3_CTS_B	GPIO1_I020	GPIO_AD_B1_04		
5	UART_RTS	TELEM_RTS	LPUART3_RTS_B	GPIO1_I021	GPIO_AD_B1_05		
6	GND						

J9 TELEM2

JST-GH 1x6, top entry · DS-009 TELEM · 3.3 V logic

PX4: TELEM2 (LPUART5)

1	5V out	5V					
2	UART_TX	TELEM2_TX	LPUART5_TX	GPIO2_I028	GPIO_B1_12		
3	UART_RX	TELEM2_RX	LPUART5_RX	GPIO2_I029	GPIO_B1_13		
4	UART_CTS	TELEM2_CTS	LPUART5_CTS_B	GPIO4_I028	GPIO_EMC_28		
5	UART_RTS	TELEM2_RTS	LPUART5_RTS_B	GPIO4_I027	GPIO_EMC_27		
6	GND						